

chapter 1: calculus review

derivative rules

the derivative of x^n is $n x^{n-1}$ **Example:** $d/dx (x^3) = 3x^2$

Important: the chain rule is essential for composite functions

Integration

the fundamental theorem of calculus says: $\int_a^b f(x) dx = F(b) - F(a)$

Definition: a definite integral computes the area under a curve between two points

Key: formulas:

- $\int x^n dx = x^{n+1} / (n+1) + C$
- $\int 1/x dx = \ln(|x|) + C$

here's a quick python function to compute numeric integration:

```
def trapezoidal_rule(f, a, b, n):  
    h = (b - a) / n
```

result = (f(a) + f(b)) / 2

```
    for i in range(1, n):  
        result += f(a + i * h)  
    return result * h
```

Note: the trapezoidal rule approximates the area under a curve **Summary:** we covered derivative rules, integration, and numerical methods

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